

Fabian Gittins

CONTACT INFORMATION	Department of Physics and Astronomy The Johns Hopkins University, 3400 N. Charles Street Baltimore, MD 21218, United States	fgittin1@jh.edu fgittins.github.io
CITIZENSHIP	United Kingdom	
RESEARCH INTERESTS	Relativistic astrophysics – Gravitational-wave astronomy – Extreme physics of neutron stars One major theme is building predictive, physically faithful neutron-star models with realistic microphysics and dynamics. Second major theme is using gravitational waves to extract this physics and constrain dense nuclear matter. Currently, advancing gravitational-wave asteroseismology to probe dense matter through modelling and detection of neutron-star oscillation modes.	
EDUCATION	PhD, Mathematics , University of Southampton, GB Advisor: Prof Nils Andersson Thesis title: <i>Gravitational waves from deformed neutron stars: mountains and tides</i> MSci, Physics , University of Birmingham, GB Grade: First class honours Undergraduate Master's degree with focus on theoretical physics	Sep 2021 Jul 2017
RESEARCH POSITIONS	Marie Skłodowska-Curie Fellow , Johns Hopkins University, US Marie Skłodowska-Curie Fellow , Utrecht University, NL Research Fellow , University of Southampton, GB PhD Researcher , University of Southampton, GB	Oct 2026–Present Oct 2024–Sep 2026 Oct 2021–Sep 2024 Sep 2017–Sep 2021
HONOURS AND AWARDS	Marie Skłodowska-Curie Postdoctoral Fellowship (Global) , EU Project lead of <i>QuarkWave</i> ; €454,103 Marie Skłodowska-Curie Postdoctoral Fellowship (European) , EU Project lead of <i>DynTideEOS</i> ; €203,464 Gravitational Physics Thesis Prize , Institute of Physics, GB Best Publication in Gravitational Physics , University of Southampton, GB Physics Scholarship , University of Birmingham, GB	Oct 2026–Sep 2029 Oct 2024–Sep 2026 2021 2021 2013

INVITED TALKS	14. <i>Physics Colloquium</i>	7 Oct 2026
	University of Maryland, Baltimore County, US	
	13. <i>Astrophysics Talks Series</i>	2 Apr 2026
	Istanbul University Observatory, TR (online)	
	12. <i>International Research Network for Nuclear Astrophysics Seminar</i>	16 Jan 2026
	(online)	
	11. <i>CoCoNuT Meeting 2025</i>	28 Oct 2025
	University of Strasbourg, FR	
	10. <i>High Energy Particle Physics and Cosmology Theory Seminar</i>	30 Sep 2025
	Johns Hopkins University, US	
	9. <i>Institute for Nuclear Theory Program 25-2b</i>	17 Sep 2025
	University of Washington, US	
	8. <i>Gravitational Wave Meeting</i>	18 Jun 2025
	National Institute for Subatomic Physics, NL (online)	
CONTRIBUTED TALKS (SELECTED)	7. <i>Astrophysics Seminar</i>	30 May 2024
	Mullard Space Science Laboratory, University College London, GB	
	6. <i>Gravitational Wave Group</i>	14 Dec 2023
	Institute of Cosmology and Gravitation, University of Portsmouth, GB	
	5. <i>SPINS-UK Seminar</i>	7 Jun 2023
	(online)	
	4. <i>Symposium on Gravitational Wave Physics and Astronomy: Genesis</i>	28 Apr 2022
	Kyoto University, JP (online)	
	3. <i>22nd BritGrav Conference</i>	5 Apr 2022
	University of Glasgow, GB (online)	
	2. <i>Colloquium</i>	6 Oct 2020
	Albert Einstein Institute, Hannover, DE (online)	
	1. <i>LIGO-Virgo Collaboration Continuous Waves Working Group</i>	5 Dec 2018
	(online)	
	27 contributed talks at 25 separate conferences and meetings, including:	
	10. <i>NNV section for (astro)particle physics fall meeting</i>	7 Nov 2025
	Soesterberg, NL	
	9. <i>Joint 24th International Conference on General Relativity and Gravitation</i>	17 Jul 2025
	and <i>16th Edoardo Amaldi Conference on Gravitational Waves</i> , Glasgow, GB	
	8. <i>XV Einstein Telescope Symposium</i>	27 May 2025
	Bologna, IT	
	7. <i>Institute for Nuclear Theory Workshop 24-89w</i>	5 Sep 2024
	University of Washington, US	
	6. <i>XIV Einstein Telescope Symposium</i>	6–7 May 2024
	Maastricht, NL	
	5. <i>SPINS-UK 2023 meeting</i>	23 Nov 2023
	Magdalen College, University of Oxford, GB	

4. *Institute for Nuclear Theory Program 22-2a* **18 Jul 2022**
University of Washington, US
3. *PHAROS Conference 2022* **18 May 2022**
La Sapienza University, Rome, IT
2. *30th Texas Symposium on Relativistic Astrophysics* **17 Dec 2019**
University of Portsmouth, GB
1. *Joint 22nd International Conference on General Relativity and Gravitation* **9 Jul 2019**
and *13th Edoardo Amaldi Conference on Gravitational Waves*, Valencia, ES

TEACHING EXPERIENCE

Lecturer, Utrecht University, NL **Apr–Jul 2026**
NS-378B, Cosmology
Third-year BSc course with ~ 70 students; mean student evaluation score of 7.6/10

Instructor, University of Southampton, GB **Feb–May 2024**
MATH1007/1009, Mathematical Methods for Physical Scientists
First-year BSc course with ~ 30 students

Guest Lecturer, University of Southampton, GB
MATH3072, Advanced Fluid Dynamics **Oct 2022, Oct 2023**
Third-year BSc course with ~ 20 students
MATH3006, Relativity, Black Holes and Cosmology **Apr 2022**
Third-year BSc course with ~ 30 students

Teaching Assistant, University of Southampton, GB **Oct 2017–May 2021**
MATH1054/1055, Mathematics for Engineering and the Environment
MATH1057, Dynamics and Relativity
MATH1058, Operational Research I and Mathematical Computing
MATH2045, Vector Calculus and Complex Variable Theory
MATH3018, Numerical Methods
MATH3087, Maths and Your Future

Teaching Assistant, King Edward’s School, Birmingham, GB **Jan–Apr 2016**
Physics (11–16 yr)

MENTORING AND SUPERVISION

PhD student mentoring

Vincent Ernst, Université Catholique de Louvain, BE **Sep 2025–Present**
Thibaut Wouters, Utrecht University, NL **Oct 2024–Present**
Rahime Matur, University of Southampton, GB **Jan 2023–Sep 2024**
Rhys Counsell, University of Southampton, GB **Sep 2021–Sep 2024**
Shanshan Yin, University of Southampton, GB **Sep 2021–Sep 2024**
Thomas Celora, University of Southampton, GB **Sep 2021–Sep 2023**

BSc student mentoring

Anouk Verburg, Utrecht University, NL

Aug 2026–Present

Tobie Walraven, Utrecht University, NL

Aug 2025–Jan 2026

**SCIENTIFIC
COLLABORATIONS****LIGO-Virgo-KAGRA Collaboration**, Member

Oct 2024–Present

LIGO Collaboration, Member

Oct 2026–Present

Virgo Collaboration, Member

Oct 2024–Sep 2026

Cosmic Explorer Consortium, Member

May 2024–Present

Einstein Telescope Collaboration, Member

Sep 2023–Present

**LEADERSHIP AND
ACADEMIC
CITIZENSHIP****International Astronomical Union**, Junior member

May 2023–Present

International Society on General Relativity and Gravitation,
Lifetime member

May 2021–Present

Institute of Physics, Member

Apr 2021–Dec 2025

Gravitational Physics Group, Committee member

Oct 2021–Sep 2025

Conference organiser**SPINS-UK 2024 meeting**, University of Southampton, GB

10–12 Sep 2024

Local organising committee, ~ 40 participants

Continuous gravitational waves and neutron stars workshop,

17–20 Jun 2024

Albert Einstein Institute, Hannover, DE

Scientific organising committee, ~ 50 participants

Gravitational Physics Annual Meeting, Institute of Physics, GB

18 Jan 2024

Scientific organising committee, ~ 50 participants

23rd BritGrav Conference, University of Southampton, GB

13–14 Apr 2023

Scientific and local organising committee, ~ 100 participants

Seminar organiser

Gravity Seminar, University of Southampton, GB

Oct 2021–Sep 2024

Weekly Gravity Reading Group, University of Southampton, GB

Jan–Jul 2021

Panellist**Fellowships Panel Discussion, GW:UK** (online)

26 Jun 2026

Funding Opportunities Workshop,

27 May 2026

Gravitational-Wave Early Career Scientists (online)**PROFESSIONAL
SERVICE****Journal referee**

Astronomy and Astrophysics, Classical and Quantum Gravity, Journal of Cosmology and

Astroparticle Physics, Journal of Physics G, Monthly Notices of the Royal Astronomical Society, Nature Astronomy, Physical Review D, Physical Review Letters, The Astrophysical Journal

Project referee

UNamur Postdoctoral Programme, University of Namur, BE **2025**
Open Fellowship, Engineering and Physical Sciences Research Council, GB **2024**

PUBLIC ENGAGEMENT

Outreach activities

Southampton Science and Engineering Festival **7 May 2022, 18 Mar 2023**
 Organised neutron-star exhibit for general public and coordinated team of 10 volunteers
 Mathematical Challenge **Mar–Apr 2020**
 Marked over 200 pupil entries
 Maths and Physics Workshop **8 Nov 2017**
 Demonstrated for ~ 100 secondary-school pupils

Press (selected)

Lightest neutron star ever found could contain compressed quarks, **24 Oct 2022**
 New Scientist
Neutron star ‘mountains’ may be blocking our view of mysterious gravitational waves, **21 Jul 2021**
 Live Science
Mountains on neutron stars are not even a millimetre tall due to extreme gravity, **21 Jul 2021**
 The Register
Scientists find tiny mountains on neutron stars that are a fraction of a millimetre tall, **19 Jul 2021**
 The Independent
Neutron Stars Have Mountains That Are Less Than a Millimeter Tall, **18 Jul 2021**
 Gizmodo
Neutron stars are remarkably smooth thanks to their intense gravity, **24 May 2021**
 New Scientist
Why don’t they just break up? Astrobites **16 Nov 2018**

COMPUTER SKILLS Advanced in Julia, Python. Intermediate in Bash, C++, Mathematica, MATLAB. Intermediate in high-performance computing (HTCondor, Slurm). Markup languages: $\LaTeX$, Markdown.

Software—Most contributions can be found at <https://github.com/fgittins>. Member of the *Bilby* development team (<https://github.com/bilby-dev/bilby>). Contributor to *SciML* (<https://sciml.ai>), in particular *NonlinearSolve.jl* (<https://github.com/SciML/NonlinearSolve.jl>). Author of *RealisticSeismology* Julia code (<https://github.com/fgittins/RealisticSeismology>).

REFERENCES

Prof Nils Andersson, Professor of Applied Mathematics

School of Mathematical Sciences

University of Southampton

University Road

Southampton, SO17 1BJ

United Kingdom

email: n.a.andersson@southampton.ac.uk

office phone: +44 23 8059 4551

Prof Chris Van Den Broeck, Professor of Physics

Institute for Gravitational and Subatomic Physics

Utrecht University

Princetonplein 1

3584 CC Utrecht

The Netherlands

email: c.f.f.vandenbroeck@uu.nl

office phone: +31 6 25 133 968

Dr David Tsang, Senior Lecturer in Physics

Department of Physics

University of Bath

Claverton Down

Bath, BA2 7AY

United Kingdom

email: d.tsang@bath.ac.uk

office phone: +44 12 2538 4539